Income Inequality, Economic Development, and Sustainability: Cross-Country Evidence on Life Expectancy and Environmental Institutions

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Executive Summary

Stakeholder: A policy analyst at an international development agency (e.g., the World Bank or UNDP) who advises regional directors on where to prioritize funding and policy support to improve both population health and environmental sustainability in low- and middle-income countries.

This project uses harmonized international data to examine how economic inequality relates to two outcomes central to global development: population health and societal sustainability. For life expectancy, we analyze 1,721 country-year observations from 2000–2023 and estimate regression models with region-specific effects to assess how the Gini index and log GDP per capita are associated with life expectancy at birth across seven World Bank regions. For societal sustainability, we study 269 country-year observations from 74 low- and lower-middle-income countries between 2005–2022 using an ordinal logistic regression of the World Bank's CPIA environmental sustainability rating on income inequality and log GDP per capita. We find that higher GDP per capita is consistently and strongly associated with longer life expectancy in all regions, while the relationship between inequality and life expectancy is weaker and highly region-dependent, with Sub-Saharan Africa standing out as the main region where higher inequality is clearly linked to shorter lives. In contrast, both higher GDP per capita and higher inequality are associated with better CPIA sustainability ratings, with GDP showing the more robust effect. These results suggest that societal economic growth is the most reliable correlate of life expectancy gains, while inequality matters for health in specific regional contexts and does not provide a simple standalone signal for sustainability in low- and lower-middle-income countries.

1. Problem Motivation

Despite substantial gains in global life expectancy over recent decades, large disparities in population health persist across and within world regions. Life expectancy at birth is widely used as a summary indicator of population health because it aggregates mortality across all ages into a single, interpretable measure of expected longevity (Deaton, 2003; Parrish, 2010). International policy frameworks further emphasize that health outcomes are closely linked to economic development, income inequality, and environmental sustainability, as reflected in the Sustainable Development Goals on health, inequality, and climate and environmental protection (United Nations, 2015). Economic inequality is often cited as a key driver of unequal outcomes, with influential work arguing that more unequal societies experience worse health and social outcomes even at similar income levels (Wilkinson & Pickett, 2009; Wilkinson, 2011), while other research stresses that the inequality–health relationship is highly context dependent and shaped by institutions and stages of development (Deaton, 2003). At the same time, international financial institutions increasingly rely on indicators such as the World Bank's Country Policy and Institutional Assessment (CPIA) environmental sustainability rating to inform concessional financing decisions for low-income countries (World Bank, 2010, 2022). For policymakers with limited resources, a central question is whether economic inequality provides useful information about where population health and environmental sustainability outcomes are most at risk, and where targeted interventions may yield the greatest joint benefits.

2. Results

2.1 Life expectancy results

2.1.1 Descriptive patterns: GDP, inequality, and life expectancy

Across all regions, countries with higher GDP per capita tend to have longer life expectancy. Figure 1 plots life expectancy against log(GDP per capita), with separate trend lines by region. The positive relationship is clear in every region, though the slope differs somewhat. In practical terms, moving from very low to middle levels of GDP per capita is associated with several additional years of expected life at birth. This confirms that economic development is a central correlate of population health in our data.

By contrast, the raw relationship between income inequality and life expectancy keeping GDP per capita constant is weaker and more mixed. Figure 2 shows life expectancy against the Gini index, again with separate regional trend lines. In many regions, the fitted lines are relatively flat or only slightly sloped, indicating that higher inequality is not systematically associated with higher or lower life expectancy on its own. This suggests that inequality is not a simple standalone predictor of population health once we look across diverse regional contexts.

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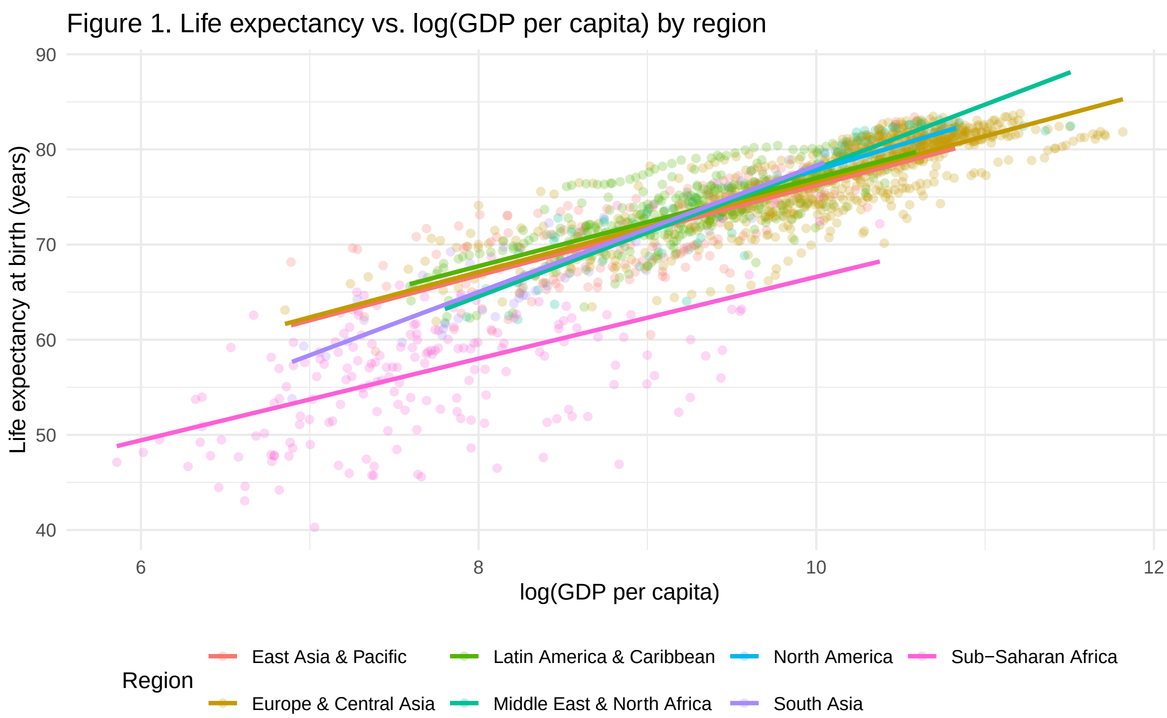

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Figure 2. Life expectancy vs. economic inequality (Gini index) by region

2.1.2 Regression results: global effects of GDP and inequality

To move beyond these descriptive patterns, we fit a linear regression model with life expectancy as the outcome and the Gini index, log(GDP per capita), region indicators, and year fixed effects as predictors. This model uses the full panel of 1,721 country–year observations and controls for broad regional differences and global time trends.

The model (Table 1) confirms that economic development is strongly and positively associated with life expectancy. Holding inequality, region, and year constant, higher log(GDP per capita) is associated with longer life expectancy, and the confidence interval for this effect is well above zero. In contrast, the global coefficient on the Gini index is much smaller in magnitude and closer to zero, indicating that, on average across regions and years, inequality is only weakly related to life expectancy once income is taken into account.

Table 1: Estimated effects of income inequality and economic development on life expectancy (year and region adjusted).

Variable	Coefficient	Lower	Upper
Gini index (per 10 points)	-0.32	-0.61	-0.03

Variable	Coefficient	Lower	Upper
log(GDP) (per unit)	4.67	4.45	4.90

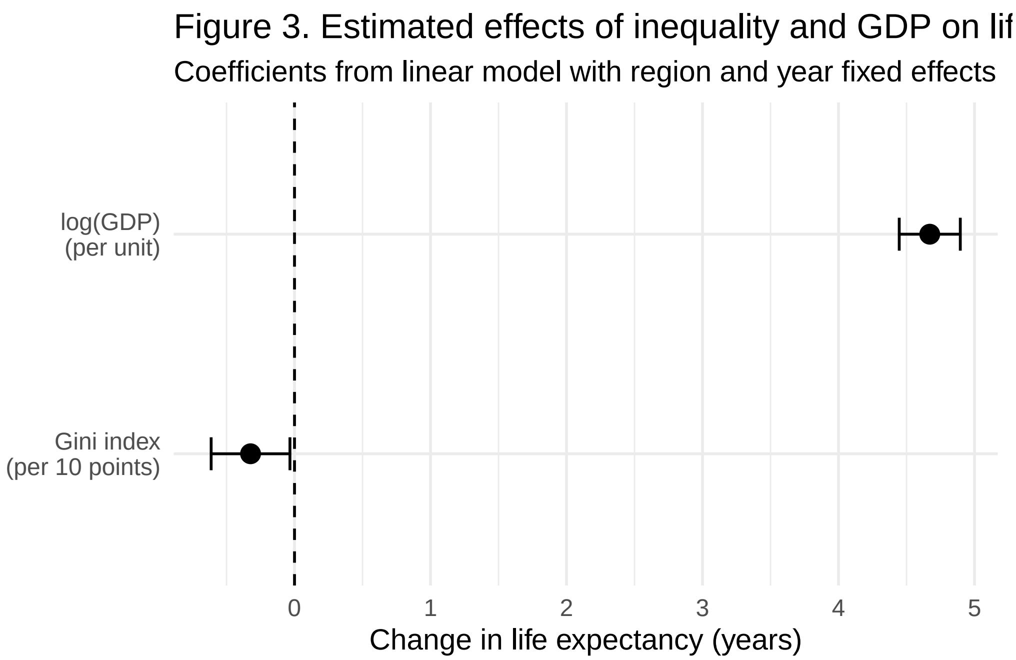


Figure 3 summarizes these regression estimates. The bar for log(GDP) sits well above zero with a relatively narrow confidence interval, indicating a robust positive effect. The bar for the Gini index is closer to zero and its interval overlaps values that are small in practice, which is consistent with the idea that inequality is not a strong global driver of life expectancy when income and time are controlled.

2.1.3 Regional differences and robustness

Although the global association between inequality and life expectancy is weak, region-specific models reveal important heterogeneity. Allowing the Gini index–life expectancy relationship to vary by region shows that Sub-Saharan Africa is the main exception, where higher inequality is clearly associated with shorter life expectancy even after controlling for GDP per capita and year. In other regions, the inequality effect is close to zero and often statistically indistinguishable from no relationship. Robustness checks indicate that these results are not driven by a small number of influential country-year observations: excluding such points does not alter the strong positive association between income and life expectancy or the distinctive role of inequality in Sub-Saharan Africa, increasing confidence in the stability of the findings.

2.2 Sustainability results:

2.2.1 Descriptive patterns: inequality, GDP, and sustainability ratings

We then focus on 269 country–year observations from low- and lower-middle-income countries with CPIA sustainability ratings. Figure 4 shows simple bivariate relationships between the sustainability rating and our two main predictors. The first panel plots sustainability against the Gini index and shows a gentle upward slope: countries with higher inequality tend to have slightly higher sustainability ratings on average, although points are widely scattered. The second panel plots sustainability against log(GDP per capita) and shows a clearer upward trend, with higher-income countries more likely to have higher sustainability ratings.

Figure 4.1. Sustainability vs. income inequality

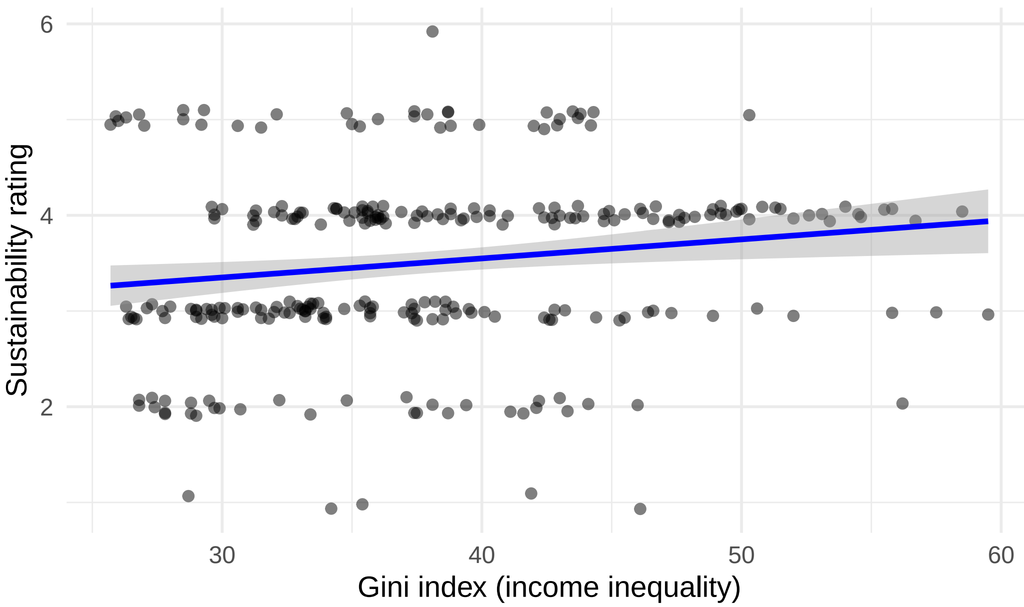


Figure 4.2. Sustainability vs. economic development

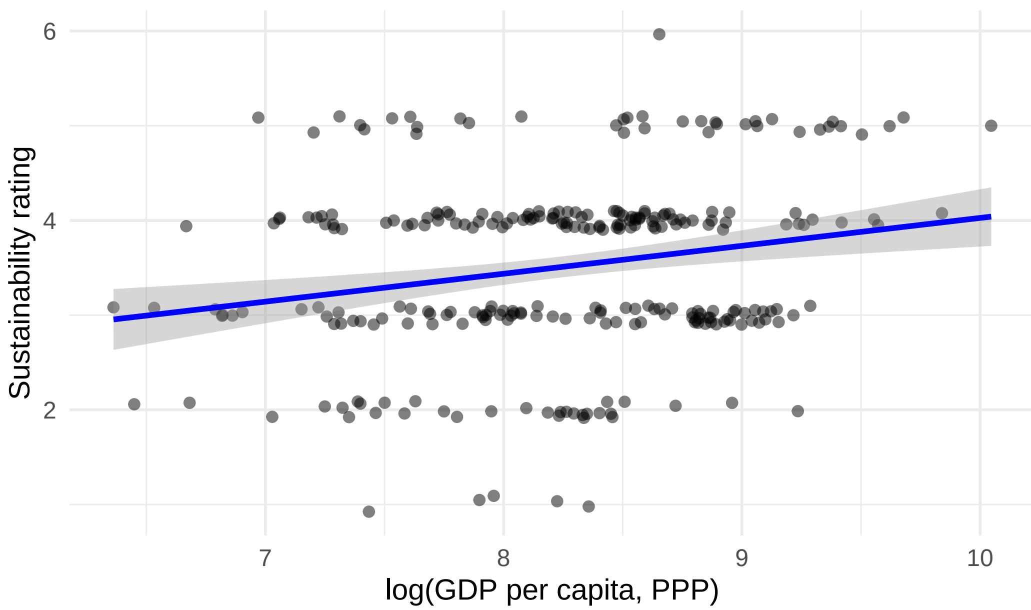


Figure 5 looks at the same relationships by sustainability category. The Gini index boxplots show that median inequality increases slightly from lower to higher sustainability categories, but there is substantial overlap in the distributions. In other words, countries with similar sustainability ratings can have quite different levels of inequality. In contrast, the log(GDP) boxplots show a clearer separation across categories, with higher-rated countries consistently having higher median GDP per capita. This suggests that economic development is a stronger differentiator of sustainability performance than inequality.

Figure 5.1. Income inequality by sustainability category

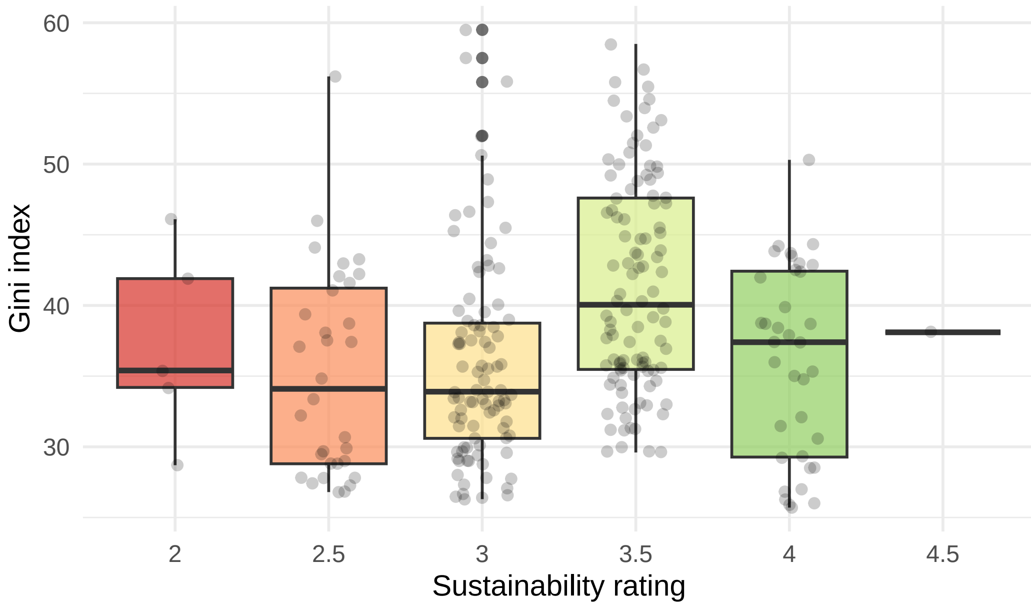
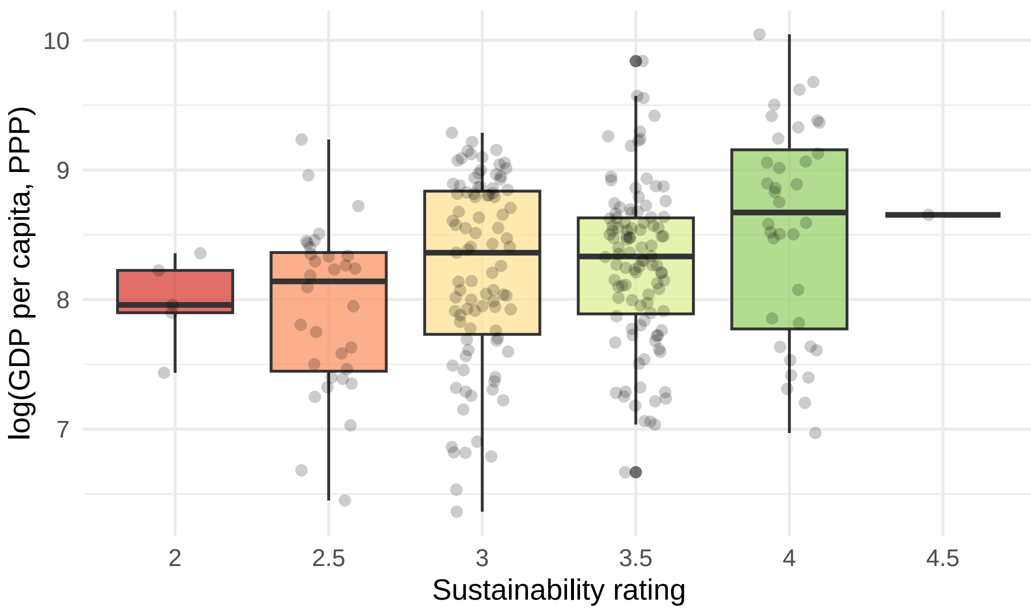


Figure 5.2. Economic development by sustainability category



2.2.2 Ordinal logistic regression results

We estimate an ordinal logistic regression with the CPIA sustainability rating as the ordered outcome and the Gini index, log GDP per capita, and year fixed effects as predictors, which accounts for the ordered nature of the score and absorbs common shocks or changes in rating practices over time. The results (Table 2) show that both inequality and income are positively associated with higher sustainability ratings. The regression coefficient for Gini index is about 0.074, implying an odds ratio of roughly 1.08 per one-point increase and approximately a doubling of the odds of a higher rating for a 10-point increase in inequality. The regression coefficient for log GDP is about 0.84, corresponding to an odds ratio of about 2.3, so a one-unit increase in log GDP per capita (roughly a 2.7-fold increase in income)is associated with more than double the odds of being in a higher sustainability category. Both associations are statistically significant.

Table 2: Ordinal logistic regression odds ratios for inequality and economic development on sustainability ratings (controlling for year).

term	estimate	std.error	statistic	conf.low	conf.high	coef.type
Gini	1.076	0.016	4.624	1.044	1.111	coefficient
log_GDP	2.319	0.183	4.598	1.625	3.334	coefficient
factor(Year)2006	1.037	0.602	0.060	0.317	3.388	coefficient
factor(Year)2007	0.988	0.582	-0.021	0.316	3.108	coefficient
factor(Year)2008	0.907	0.649	-0.151	0.253	3.251	coefficient
factor(Year)2009	1.111	0.587	0.179	0.350	3.521	coefficient
factor(Year)2010	1.128	0.553	0.217	0.381	3.351	coefficient
factor(Year)2011	0.609	0.575	-0.861	0.197	1.889	coefficient
factor(Year)2012	0.952	0.589	-0.083	0.300	3.035	coefficient
factor(Year)2013	0.658	0.659	-0.635	0.180	2.410	coefficient
factor(Year)2014	1.072	0.620	0.112	0.317	3.632	coefficient
factor(Year)2015	1.151	0.587	0.240	0.364	3.651	coefficient
factor(Year)2016	2.257	0.670	1.215	0.608	8.495	coefficient
factor(Year)2017	0.999	0.819	-0.001	0.201	5.047	coefficient
factor(Year)2018	2.154	0.582	1.319	0.690	6.794	coefficient
factor(Year)2019	2.551	0.638	1.469	0.733	9.014	coefficient
factor(Year)2020	3.584	0.861	1.482	0.656	19.903	coefficient
factor(Year)2021	1.869	0.688	0.909	0.487	7.289	coefficient
factor(Year)2022	1.200	0.805	0.226	0.247	5.936	coefficient

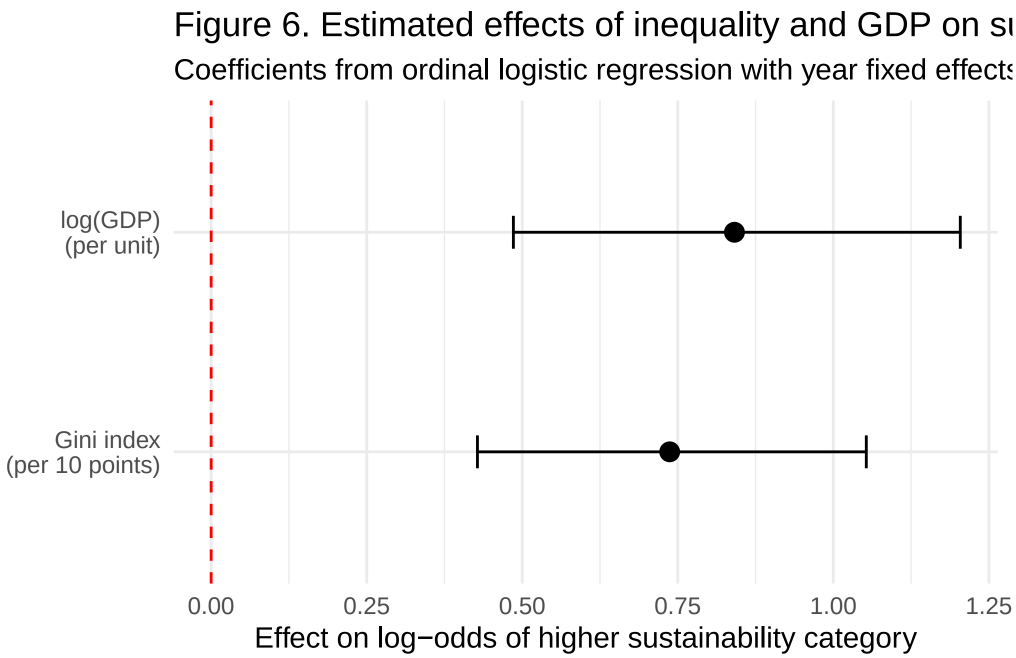


Figure 6 compares the regression coefficients for Gini index and log(GDP) on the log-odds scale. Both bars are clearly above zero, but the bar for log(GDP) is taller, reflecting its larger affect with sustainability ratings. This supports the idea that among low- and lower-middle-income countries, economic development holds more weight in predicting institutional sustainability than income distribution, even though inequality also has a positive association.

Year fixed effects are generally small and imprecise, and the likelihood ratio test shows that including Gini index, log(GDP), and year jointly improves model fit over a model with only intercepts. McFadden's pseudo-R^2 is modest (around 0.06), which is typical for institutional rating data and reflects the fact that sustainability outcomes depend on many unobserved country-specific factors. For our purposes, the key point is that the estimated associations for Gini index and GDP are statistically robust and directionally clear.

2.2.3 Counterintuitive inequality effect

The positive coefficient on the Gini index is counterintuitive, as higher inequality is often expected to be associated with less sustainable policies and institutions. Several factors may explain this result. CPIA ratings are observed only for IDA-eligible low- and lower-middle-income countries, within which higher inequality may coincide with greater urbanization, more diversified economies, or stronger administrative and statistical capacity that support environmental policy frameworks. In addition, both inequality and sustainability ratings may reflect deeper, unobserved structural characteristics such as governance quality, and CPIA scores themselves emphasize formal policies and institutions rather than distributional outcomes.

The practical implication is that inequality alone is not a reliable screening indicator for environmental sustainability performance in this sample. GDP per capita shows a stronger and more consistent association with sustainability ratings, and the positive inequality effect should be interpreted cautiously and examined further rather than viewed as evidence that higher inequality improves environmental outcomes.

2.3 Integrated implications

Taken together, the two analyses reveal a consistent pattern. Economic development, measured by log GDP per capita, is the dominant predictor of both outcomes. Higher income levels are strongly associated with longer life expectancy across all regions and with higher sustainability ratings among low- and lower-middle-income countries. Economic inequality plays a more limited and context-dependent role. Its association with life expectancy is generally weak and imprecise, except in Sub-Saharan Africa, where higher inequality is clearly linked to shorter life expectancies. For sustainability, the positive association with inequality is likely driven by sample restrictions, measurement issues, and unobserved institutional factors rather than a causal relationship.

For a regional policy advisor, these results suggest several practical lessons. Policies that raise economic productivity remain the most reliable lever for improving population health and supporting environmental institutions. Inequality warrants targeted attention in specific contexts, particularly in Sub-Saharan Africa, but it should not be used in isolation as a proxy for environmental sustainability performance. More broadly, because both health and sustainability outcomes are shaped by complex and partially unobserved factors, cross-country statistical results should be complemented with country-specific institutional knowledge when setting priorities and designing interventions.

3. Methodology

Our analysis uses regression models as descriptive tools to examine how economic inequality and economic development are associated with life expectancy and sustainability. In each case, we control for observed factors such as year and region so that estimated coefficients can be interpreted as associations holding those factors constant, rather than as causal effects. The models are designed to provide policy-relevant summaries of how differences in income levels and income distribution tend to align with differences in outcomes across countries and over time.

3.1 Models for life expectancy

To study life expectancy at birth, measured in years, we estimate linear regression models in which life expectancy is explained by the Gini index, log GDP per capita (PPP), region, and year. The coefficient on log GDP per capita indicates how many additional years of life expectancy are associated with a one-unit increase in log income, which corresponds roughly to a 2.7-fold increase in GDP, holding inequality, region, and year fixed. The coefficient on the Gini index captures how life expectancy changes as inequality rises after controlling for income and other factors, and we rescale this effect to reflect a 10-point increase in the Gini index for easier interpretation. Because the relationship between inequality and health may differ across regions, we also estimate a specification that allows the effects of inequality and income to vary by region, producing region-specific slopes that highlight where inequality is more strongly linked to shorter or longer lives.

3.2 Models for sustainability ratings

For sustainability, the outcome variable is the CPIA environmental sustainability rating, which is an ordered categorical measure, so we estimate an ordinal logistic regression rather than a linear model. The CPIA sustainability rating is a measure developed as part of the Country Policy and Institutional Assessment (CPIA) as an indicator of a country's efforts via policies and institutions for sustainable development. This can include natural resources management, pollution, economic growth, and poverty reduction measures. The predictors in this model are the Gini index, log GDP per capita, and year, and the coefficients describe how these variables shift the log odds of a country receiving a higher rather than a lower sustainability rating, holding other factors constant. Interpreting the exponentiated coefficients as odds ratios, a value greater than one indicates higher odds of being in a higher sustainability category; for example, an odds ratio of about 1.08 for the Gini index implies that a one-point increase in inequality is associated with an 8 percent increase in the odds of a higher rating, with larger changes implied for a 10-point increase. Year indicators absorb common global trends or changes in scoring standards, allowing the analysis to focus on how cross-country differences in income and inequality align with differences in environmental sustainability performance.

4. Recommended Next Steps

Based on our results and the limitations of the current data, we see three concrete next steps for the stakeholder.

4.1. Prioritize societal economic growth as the most reliable lever

Economic development is consistently associated with longer life expectancy and stronger environmental institutions, supporting continued focus on investments in basic infrastructure, primary health systems, water and sanitation, energy access, and productivity, alongside monitoring how growth translates into health and sustainability gains over time.

4.2. Treat inequality as a targeted risk signal, not a universal trigger

Inequality warrants focused attention in contexts such as Sub-Saharan Africa, where it is closely linked to poorer health outcomes, but it should not be used in isolation as a proxy for sustainability or average health performance in other regions without supporting country-level evidence.

4.3. Strengthen data and diagnostics before linking finance to scores

Given measurement limitations in life expectancy, inequality, and CPIA sustainability ratings, financing decisions should be supported by improved statistical capacity, complementary objective indicators, richer covariates, and targeted country case work that combines global model signals with local institutional knowledge. Specifically, the dataset used for CPIA sustainability ratings was missing data for many countries, including but not limited to many higher-income countries, and the dataset used for Gini index was also missing data for a large portions of countries globally.

Appendix A. Data

A.1 Data sources

This project builds two country–year panels from international data:

1. A health panel for Research Question 1, linking income inequality, economic development, and life expectancy at birth.
2. A sustainability panel for Research Question 2, linking inequality, economic development, and CPIA sustainability ratings.

All data are drawn from publicly available international sources:

- Economic inequality (Gini index): UN/World Bank inequality series, downloaded from UNdata (https://data.un.org/Data.aspx?q=gini&d=WDI&f=Indicator_Code%3aSI.POV.GINI).
- GDP per capita (PPP, constant international dollars): World Bank World Development Indicators series NY.GDP.PCAP.PP.CD, accessed via UNdata (https://data.un.org/Data.aspx?q=gdp+per+capita&d=WDI&f=Indicator_Code%3aNY.GDP.PCAP.PP
- Life expectancy at birth (years): UN life table data from UNdata, using the "Medium" variant (https://data.un.org/Data.aspx?q=life+expectancy&d=PopDiv&f=variableID%3a68).
- Sustainability ratings: World Bank Country Policy and Institutional Assessment (CPIA), sustainability criterion, pre-processed into a country–year dataset, taken from the UN website (https://data.un.org/Data.aspx?q=sustainability&d=WDI&f=Indicator_Code%3aIQ.CPA.ENVI

The Table below reads the raw (or pre-cleaned) files and reports the row counts for each source so that the scope of the input data is transparent

Table 3: Row counts for raw and pre-cleaned data sources.

dataset	n_rows
UN Gini (RQ1)	2111
UN GDP (RQ1)	8268
UN Life expectancy (RQ1)	84360
UN Gini (RQ2)	2111
UN GDP (RQ2)	5609
CPIA sustainability (RQ2)	1437

A.2 Life expectancy panel

For the first research question, we construct a country–year panel where each observation has life expectancy at birth, Gini, and GDP per capita for the same country and year.

Cleaning decisions that affect the model:

1. Country alignment: when cleaning GDP and life expectancy, we keep only countries that appear in the Gini dataset.
2. Remove missing GDP and life expectancy: rows with missing GDP or life expectancy are dropped before merging.
3. Merge on Country and Year: we use inner joins so that only overlapping country–years are kept.
4. Drop remaining missing values in any of the three core variables after merging.
5. Assign regions and drop rows without a valid region (e.g., aggregates).
6. Transform GDP: we use log(GDP) in the models rather than raw GDP.

Table 4: Missing values in LifeExp, Gini, and GDP before final filtering.

n_rows_total	miss_LifeExp	miss_Gini	miss_GDP
1721	0	0	0

Table 5: Final life expectancy analysis sample: countries, years, and observations.

n_countries	n_years	n_rows
152	34	1721

A.3 Sustainability panel

A.3.1 Cleaning and merging

For the sustainability panel, we construct a sustainability panel where each observation has a CPIA sustainability rating, Gini, and GDP per capita for the same country and year. CPIA scores are only reported for IDA-eligible low and lower-middle-income countries, so the panel is restricted to that group from the outset.

Cleaning decisions that affect the model:

1. Harmonize column names in the sustainability file.
2. Inner join on Country and Year across sustainability, Gini, and GDP.
3. Check for missing values in the three analysis variables.
4. Create ordered outcome and log(GDP); drop any rows with non-finite log(GDP) (none in practice).

Table 6: Missing values in sustainability, Gini, and GDP.

Total Obs	Missing Sustainability	Missing Gini	Missing GDP	Missing Any Variable
269	0	0	0	0

Table 7: Final sustainability analysis sample: countries, years, and observations.

n_countries	n_years	n_rows
74	18	269

We can also show how CPIA ratings are distributed in the final sample from the table A.6 below.

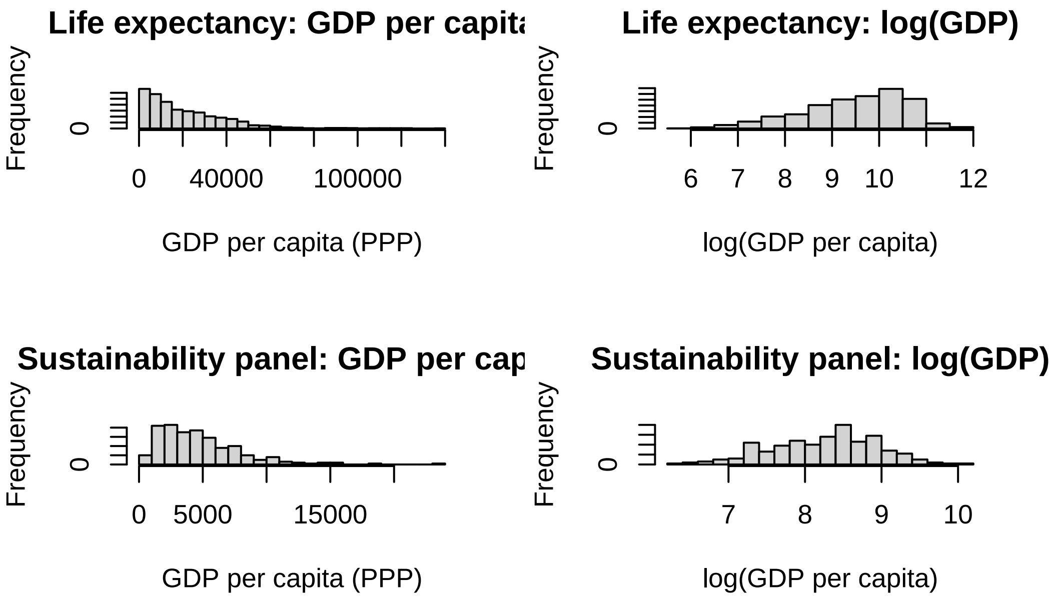
Table 8: Distribution of CPIA environmental sustainability ratings.

Sustainability level	Count
2	5
2.5	32
3	91
3.5	104
4	36

Sustainability level	Count
4.5	1

A.4 Unified illustration of GDP transformation

Both panels use **log(GDP per capita)** instead of raw GDP in the final models. To keep the appendix unified, we use a single figure that shows the distribution of GDP and log(GDP) in **both** samples.



Appendix B. Models for Life Expectancy Panel

B.1 Modelling approach and variables

The outcome for Research Question 1 is life expectancy at birth (years), a continuous variable. Since we want to relate life expectancy to income inequality and economic development while controlling for time, we use linear regression models with:

- Country–year observations
- Year fixed effects (factor for calendar year)
- Cluster-robust standard errors at the country level

Main predictors:

- Gini: Gini index of income inequality (continuous)
- log_GDP: natural log of GDP per capita (PPP, constant international dollars)
- Region: world region from countrycode (factor)

We fit two main models:

1. **Baseline panel model:**

$$LifeExp_{ct} = \beta_0 + \beta_1 Gini_{ct} + \beta_2 log(GDP_{ct}) + \gamma_t + \varepsilon_{ct}$$

2. **Region-interaction model:**

$$LifeExp_{ct} = (\beta_0 + \beta_{0r}) + (\beta_1 + \beta_{1r})Gini_{ct} + (\beta_2 + \beta_{2r})log(GDP_{ct}) + \gamma_t + \varepsilon_{ct}$$

where r indexes regions, and $ _t$ are year effects.

Assumptions (linear models):

- Linearity between predictors and life expectancy
- Additivity of effects within each region specification
- Homoskedasticity and approximate normality of residuals (checked with plots)
- Independence of errors within country–year pairs, with clustering at the country level to account for repeated observations over time
- No omitted variables that are strongly correlated with both predictors and the outcome after controlling for year and region

B.2 Baseline panel model

Table 9: Baseline linear panel model of life expectancy on inequality, log(GDP), and year fixed effects (cluster-robust standard errors).

	term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	(Intercept)	13.051	3.580	3.645	0.000	6.033	20.068
Gini	Gini	0.018	0.041	0.447	0.655	-0.062	0.098
log_GDP	log_GDP	6.529	0.322	20.304	0.000	5.899	7.159
Year_f1991	Year_f1991	-3.382	1.245	-2.716	0.007	-5.822	-0.942
Year_f1992	Year_f1992	-3.068	1.158	-2.650	0.008	-5.337	-0.798

	term	estimate	std.error	statistic	p.value	conf.low	conf.high
Year_f1993	Year_f1993	-2.878	1.015	-2.836	0.005	-4.866	-0.889
Year_f1994	Year_f1994	-3.165	1.092	-2.897	0.004	-5.306	-1.024
Year_f1995	Year_f1995	-1.850	0.862	-2.147	0.032	-3.540	-0.161
Year_f1996	Year_f1996	-2.340	1.041	-2.249	0.025	-4.380	-0.300
Year_f1997	Year_f1997	-1.400	1.096	-1.277	0.202	-3.548	0.749
Year_f1998	Year_f1998	-2.696	1.014	-2.658	0.008	-4.683	-0.708
Year_f1999	Year_f1999	-0.743	0.810	-0.918	0.359	-2.331	0.844
Year_f2000	Year_f2000	-2.311	1.099	-2.102	0.036	-4.466	-0.156
Year_f2001	Year_f2001	-1.296	1.008	-1.286	0.199	-3.272	0.679
Year_f2002	Year_f2002	-2.195	1.061	-2.069	0.039	-4.275	-0.115
Year_f2003	Year_f2003	-2.187	1.064	-2.055	0.040	-4.273	-0.101
Year_f2004	Year_f2004	-1.525	0.955	-1.596	0.111	-3.397	0.347
Year_f2005	Year_f2005	-2.050	0.955	-2.146	0.032	-3.922	-0.178
Year_f2006	Year_f2006	-2.100	0.952	-2.206	0.028	-3.967	-0.234
Year_f2007	Year_f2007	-2.309	0.957	-2.412	0.016	-4.185	-0.433
Year_f2008	Year_f2008	-2.736	1.021	-2.679	0.007	-4.738	-0.734
Year_f2009	Year_f2009	-2.615	1.049	-2.493	0.013	-4.671	-0.560
Year_f2010	Year_f2010	-1.944	0.983	-1.978	0.048	-3.870	-0.017
Year_f2011	Year_f2011	-2.310	0.944	-2.446	0.015	-4.161	-0.459
Year_f2012	Year_f2012	-2.240	0.984	-2.276	0.023	-4.169	-0.311
Year_f2013	Year_f2013	-1.788	0.953	-1.877	0.061	-3.655	0.079
Year_f2014	Year_f2014	-1.963	0.957	-2.051	0.040	-3.838	-0.087
Year_f2015	Year_f2015	-2.442	1.020	-2.394	0.017	-4.441	-0.443
Year_f2016	Year_f2016	-1.936	0.993	-1.950	0.051	-3.883	0.010
Year_f2017	Year_f2017	-2.743	1.004	-2.731	0.006	-4.711	-0.774
Year_f2018	Year_f2018	-3.106	1.024	-3.035	0.002	-5.112	-1.100
Year_f2019	Year_f2019	-2.376	0.976	-2.434	0.015	-4.290	-0.463
Year_f2020	Year_f2020	-3.465	1.003	-3.454	0.001	-5.432	-1.499
Year_f2021	Year_f2021	-5.304	1.005	-5.278	0.000	-7.274	-3.334
Year_f2022	Year_f2022	-2.992	1.088	-2.751	0.006	-5.123	-0.860
Year_f2023	Year_f2023	-2.174	1.461	-1.488	0.137	-5.037	0.689

Brief Interpretation

- The coefficient on log_GDP is positive and highly significant. Higher GDP per capita is associated with higher life expectancy, holding inequality and year constant.
- The coefficient on Gini index is small and negative in this specification. Once we control for GDP per capita and year, higher inequality is weakly associated with lower life expectancy.

B.3 Region-interaction model

Table 10: Linear model of life expectancy with interactions between inequality, log(GDP), and world region (cluster-robust standard errors).

	term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	(Intercept)	25.857	7.746	3.338	0.001	10.675	41.039
Gini	Gini	0.118	0.103	1.142	0.254	-0.084	0.320
log_GDP	log_GDP	4.612	0.765	6.029	0.000	3.113	6.112
RegionEurope & Central Asia	RegionEurope & Central Asia	1.649	10.029	0.164	0.869	-18.008	21.305
RegionLatin America & Caribbean	RegionLatin America & Caribbean	7.308	10.373	0.704	0.481	-13.023	27.639
RegionMiddle East & North Africa	RegionMiddle East & North Africa	-20.248	10.017	-2.021	0.043	-39.880	-0.616
RegionNorth America	RegionNorth America	11.437	10.587	1.080	0.280	-9.313	32.187
RegionSouth Asia	RegionSouth Asia	-19.097	9.737	-1.961	0.050	-38.182	-0.013
RegionSub-Saharan Africa	RegionSub-Saharan Africa	6.960	8.771	0.794	0.428	-10.230	24.151
Year_f1991	Year_f1991	-1.055	0.844	-1.250	0.211	-2.710	0.599
Year_f1992	Year_f1992	-1.716	0.914	-1.878	0.061	-3.507	0.075
Year_f1993	Year_f1993	-0.532	0.654	-0.814	0.416	-1.813	0.749
Year_f1994	Year_f1994	-0.388	0.880	-0.440	0.660	-2.113	1.338
Year_f1995	Year_f1995	-0.775	0.690	-1.124	0.261	-2.128	0.577
Year_f1996	Year_f1996	-1.297	0.737	-1.761	0.078	-2.741	0.146
Year_f1997	Year_f1997	-0.153	0.750	-0.205	0.838	-1.623	1.316
Year_f1998	Year_f1998	-1.800	0.761	-2.366	0.018	-3.291	-0.309
Year_f1999	Year_f1999	-0.294	0.667	-0.441	0.659	-1.602	1.014
Year_f2000	Year_f2000	-0.537	0.796	-0.674	0.500	-2.097	1.024

	term	estimate	std.error	statistic	p.value	conf.low	conf.high
Year_f2001	Year_f2001	-0.076	0.884	-0.086	0.931	-1.808	1.656
Year_f2002	Year_f2002	-0.809	0.792	-1.021	0.307	-2.362	0.744
Year_f2003	Year_f2003	-0.720	0.772	-0.933	0.351	-2.232	0.793
Year_f2004	Year_f2004	0.034	0.767	0.044	0.965	-1.469	1.536
Year_f2005	Year_f2005	0.076	0.752	0.101	0.919	-1.397	1.550
Year_f2006	Year_f2006	-0.387	0.798	-0.485	0.628	-1.950	1.176
Year_f2007	Year_f2007	-0.424	0.815	-0.520	0.603	-2.021	1.174
Year_f2008	Year_f2008	-0.402	0.822	-0.489	0.625	-2.014	1.210
Year_f2009	Year_f2009	-0.365	0.807	-0.453	0.651	-1.948	1.217
Year_f2010	Year_f2010	0.388	0.817	0.475	0.635	-1.213	1.990
Year_f2011	Year_f2011	0.030	0.800	0.037	0.970	-1.538	1.598
Year_f2012	Year_f2012	-0.197	0.844	-0.234	0.815	-1.851	1.457
Year_f2013	Year_f2013	0.324	0.839	0.386	0.700	-1.321	1.968
Year_f2014	Year_f2014	0.595	0.833	0.714	0.475	-1.038	2.227
Year_f2015	Year_f2015	0.363	0.815	0.445	0.656	-1.235	1.961
Year_f2016	Year_f2016	0.636	0.882	0.721	0.471	-1.092	2.364
Year_f2017	Year_f2017	-0.206	0.855	-0.241	0.810	-1.881	1.469
Year_f2018	Year_f2018	-0.168	0.874	-0.192	0.848	-1.880	1.545
Year_f2019	Year_f2019	0.256	0.913	0.280	0.780	-1.534	2.045
Year_f2020	Year_f2020	-0.870	0.895	-0.972	0.331	-2.624	0.885

	term	estimate	std.error	statistic	p.value	conf.low	conf.high
Year_f2021	Year_f2021	-2.177	0.939	-2.320	0.020	-4.017	-0.338
Year_f2022	Year_f2022	-0.690	1.095	-0.631	0.528	-2.836	1.455
Year_f2023	Year_f2023	0.113	1.316	0.086	0.931	-2.467	2.693
Gini:RegionEurope & Central Asia	Gini:RegionEurope & Central Asia	-0.066	0.127	-0.520	0.603	-0.314	0.182
Gini:RegionLatin America & Caribbean	Gini:RegionLatin America & Caribbean	-0.120	0.125	-0.961	0.337	-0.365	0.125
Gini:RegionMiddle East & North Africa	Gini:RegionMiddle East & North Africa	0.047	0.146	0.324	0.746	-0.239	0.333
Gini:RegionNorth America	Gini:RegionNorth America	-0.066	0.154	-0.428	0.668	-0.367	0.235
Gini:RegionSouth Asia	Gini:RegionSouth Asia	0.158	0.165	0.959	0.338	-0.165	0.480
Gini:RegionSub-Saharan Africa	Gini:RegionSub-Saharan Africa	-0.364	0.115	-3.155	0.002	-0.590	-0.138
log_GDP:RegionEurope & Central Asia	log_GDP:RegionEurope & Central Asia	0.160	0.878	0.182	0.856	-1.562	1.882
log_GDP:RegionLatin America & Caribbean	log_GDP:RegionLatin America & Caribbean	-0.198	0.860	-0.230	0.818	-1.884	1.488
log_GDP:RegionMiddle East & North Africa	log_GDP:RegionMiddle East & North Africa	2.069	0.905	2.287	0.022	0.295	3.842
log_GDP:RegionNorth America	log_GDP:RegionNorth America	-0.643	0.944	-0.681	0.496	-2.493	1.208
log_GDP:RegionSouth Asia	log_GDP:RegionSouth Asia	1.557	0.865	1.800	0.072	-0.139	3.252
log_GDP:RegionSub-Saharan Africa	log_GDP:RegionSub-Saharan Africa	-0.053	0.971	-0.055	0.956	-1.957	1.851

Brief Interpretation

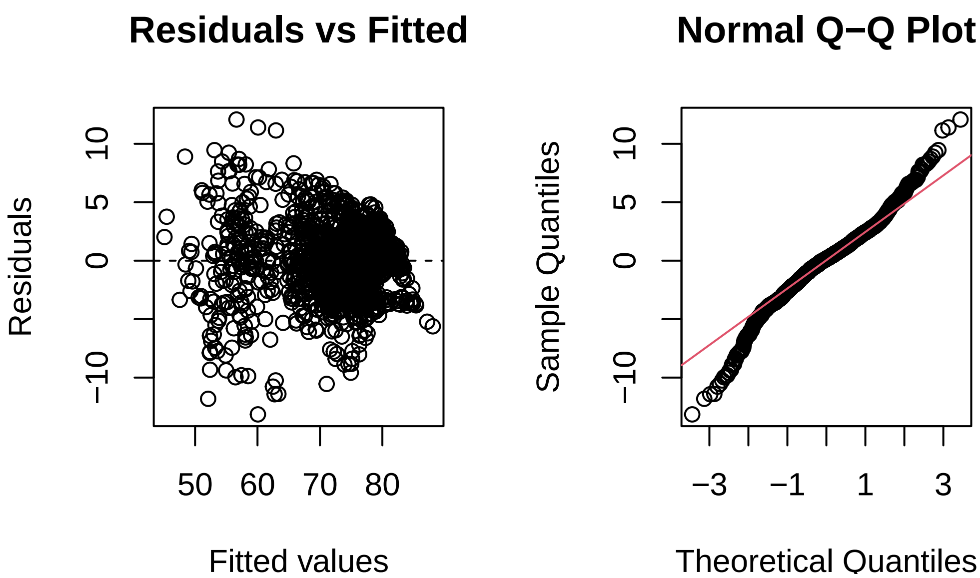
- The main Gini index and log_GDP coefficients correspond to the reference region (for example, East Asia & Pacific).
- Interaction terms like *Gini:RegionSub-Saharan Africa* show how the inequality–life expectancy slope differs in that region compared to the reference.
- Results indicate that the association between inequality and life expectancy varies by region, satisfying the requirement to use at least one interaction term and providing more nuanced policy insight.

B.4 Model assessment and changes

We assessed model fit and assumptions using residual diagnostics and influence checks.

1. **GDP transformation decision** Residual plots showed non-linearity when using raw GDP but a more stable pattern after log-transforming GDP. That motivated the use of log(GDP) in all final models instead of raw GDP.

2. **Residual diagnostics for interaction model**



Brief Interpretation

- Residuals versus fitted values show no strong curvature or funnel shape.
- The Q–Q plot shows some deviation in the tails, but not enough to overturn the linear model conclusions.

3. **Influential observations**

```
Number of potentially influential country-year observations: 157
```

Brief Interpretation:

Refitting the interaction model without these high-leverage points produced similar coefficients for Gini index and log(GDP), so we retained the full-sample interaction model as the primary specification.

Appendix C. Models for Research Question 2

C.1 Modelling approach and variables

The outcome for Research Question 2 is the CPIA sustainability rating, which is:

- Ordered categories (2.0, 2.5, 3.0, 3.5, 4.0, 4.5)
- Bounded and discrete, not continuous

Because of this, we use an **ordinal logistic regression (proportional odds model)** instead of a linear model.

Main predictors:

- Gini: Gini index, economic inequality
- log_GDP: natural log of GDP per capita (PPP, constant)
- factor(Year): year fixed effects to capture time trends or changes in CPIA scoring

We fitted the model as below:

$$logitP(Y \leq j) = \alpha_j \ - \beta_1 \ \cdot Gini - \beta_2 \ \cdot log(GDP) - \gamma_{Year}$$

where Y is the sustainability rating, j indexes the thresholds, and negative signs indicate that higher predictor values increase the odds of being in higher sustainability categories.

Assumptions (ordinal logistic model):

- Proportional odds: the effect of each predictor is constant across all cumulative splits of the outcome.
- Independence of observations (country–year pairs) conditional on predictors.
- Correct functional form for Gini and log(GDP) (checked with scatterplots and transformation of GDP).
- No strong unmeasured confounders that are correlated with both inequality, GDP, and sustainability after conditioning on year.

C.2 Ordinal logistic regression model

Table 11: Ordinal logistic regression coefficients (log-odds scale) for sustainability ratings on inequality, log(GDP), and year.

term	estimate	std.error	statistic	conf.low	conf.high	coef.type
Gini	0.074	0.016	4.624	0.043	0.105	coefficient
log_GDP	0.841	0.183	4.598	0.486	1.204	coefficient
factor(Year)2006	0.036	0.602	0.060	-1.149	1.220	coefficient
factor(Year)2007	-0.012	0.582	-0.021	-1.153	1.134	coefficient
factor(Year)2008	-0.098	0.649	-0.151	-1.374	1.179	coefficient
factor(Year)2009	0.105	0.587	0.179	-1.051	1.259	coefficient
factor(Year)2010	0.120	0.553	0.217	-0.966	1.209	coefficient
factor(Year)2011	-0.495	0.575	-0.861	-1.626	0.636	coefficient
factor(Year)2012	-0.049	0.589	-0.083	-1.204	1.110	coefficient
factor(Year)2013	-0.418	0.659	-0.635	-1.713	0.880	coefficient
factor(Year)2014	0.069	0.620	0.112	-1.149	1.290	coefficient
factor(Year)2015	0.141	0.587	0.240	-1.011	1.295	coefficient
factor(Year)2016	0.814	0.670	1.215	-0.498	2.139	coefficient
factor(Year)2017	-0.001	0.819	-0.001	-1.604	1.619	coefficient
factor(Year)2018	0.767	0.582	1.319	-0.371	1.916	coefficient
factor(Year)2019	0.937	0.638	1.469	-0.311	2.199	coefficient
factor(Year)2020	1.277	0.861	1.482	-0.422	2.991	coefficient
factor(Year)2021	0.625	0.688	0.909	-0.720	1.986	coefficient
factor(Year)2022	0.182	0.805	0.226	-1.397	1.781	coefficient

C.3 Odds ratios on the exponential scale

Table 12: Table C.2. Ordinal logistic regression odds ratios for inequality and economic development on sustainability ratings (controlling for year).

term	estimate	std.error	statistic	conf.low	conf.high	coef.type
Gini	1.076	0.016	4.624	1.044	1.111	coefficient
log_GDP	2.319	0.183	4.598	1.625	3.334	coefficient
factor(Year)2006	1.037	0.602	0.060	0.317	3.388	coefficient
factor(Year)2007	0.988	0.582	-0.021	0.316	3.108	coefficient
factor(Year)2008	0.907	0.649	-0.151	0.253	3.251	coefficient
factor(Year)2009	1.111	0.587	0.179	0.350	3.521	coefficient
factor(Year)2010	1.128	0.553	0.217	0.381	3.351	coefficient
factor(Year)2011	0.609	0.575	-0.861	0.197	1.889	coefficient
factor(Year)2012	0.952	0.589	-0.083	0.300	3.035	coefficient
factor(Year)2013	0.658	0.659	-0.635	0.180	2.410	coefficient

term	estimate	std.error	statistic	conf.low	conf.high	coef.type
factor(Year)2014	1.072	0.620	0.112	0.317	3.632	coefficient
factor(Year)2015	1.151	0.587	0.240	0.364	3.651	coefficient
factor(Year)2016	2.257	0.670	1.215	0.608	8.495	coefficient
factor(Year)2017	0.999	0.819	-0.001	0.201	5.047	coefficient
factor(Year)2018	2.154	0.582	1.319	0.690	6.794	coefficient
factor(Year)2019	2.551	0.638	1.469	0.733	9.014	coefficient
factor(Year)2020	3.584	0.861	1.482	0.656	19.903	coefficient
factor(Year)2021	1.869	0.688	0.909	0.487	7.289	coefficient
factor(Year)2022	1.200	0.805	0.226	0.247	5.936	coefficient

Brief Interpretation:

- The Gini index odds ratio > 1 implies that, within this sample of low and lower-middle-income countries, higher income inequality is associated with higher odds of being in a higher CPIA sustainability category.
- The log_GDP odds ratio > 1 is larger in magnitude, indicating that economic development has a stronger association with sustainability ratings than inequality.
- Year coefficients capture shifts in CPIA scoring or global environmental policy trends over time.

C.4 Model assessment and fit

We evaluated overall model fit and compared the full model to a null model with only intercepts.

Table 13: Deviance and AIC for null and full ordinal logistic models.

model	deviance	AIC
Null	727.03	737.03
Full	683.38	731.38

```
Likelihood ratio test (full vs null):

  Chi-square statistic: 43.65

  Degrees of freedom: 19
```

```
  p-value: 0.0010558
```

```
McFadden pseudo R^2: 0.06
```

Brief Interpretation

- The likelihood ratio test strongly favors the full model over the null model.
- McFadden's pseudo R^2 is in the typical range for ordinal logistic models, indicating a meaningful but not complete explanation of sustainability ratings.

Other Tests

Furthermore, we wanted to test for any leverage points and collinearity using the Variance Inflation Factor (VIF).

```
Number of observations with large prediction errors (|residual| >= 1.5 categories): 23



Table: Coefficient Stability Analysis: Full vs Reduced Model

|Predictor | Full_Model| Reduced_Model| Difference| Pct_Change|
|:---------|----------:|-------------:|----------:|----------:|
|Gini      |     0.0736|        0.1088|    -0.0351|      47.72|
|log_GDP   |     0.8411|        1.1345|    -0.2934|      34.88|

Conclusion: Maximum coefficient change is 47.72 %. Some sensitivity to influential

        observations exists, but main conclusions remain valid.

Variance Inflation Factors (VIF):

Continuous Predictors:

                 GVIF Df GVIF^(1/(2*Df))
Gini         1.141925  1        1.068609
log_GDP      1.192224  1        1.091890
factor(Year) 1.133485 17        1.003692
```

First, we evaluated the effect of influential observations in the data: Twenty-three countries (9% of the sample) have large prediction errors, and removing them increases the estimated effects by 35-48%. Despite this sensitivity, the main conclusion holds: both inequality and GDP per capita are positively linked to sustainability ratings, with GDP being the stronger predictor. Second, we evaluated VIF scores: all VIF values for continuous predictors are well below the conventional threshold of 5, indicating no multicollinearity concerns. The low VIF values (around 1.1-1.2) suggest that Gini and log(GDP) are nearly orthogonal, meaning they capture distinct aspects of country characteristics with minimal overlap.

C.5 Graphical checks and proportional odds discussion

we also used graphs to understand the relationships between predictors and the ordinal outcome.

Figure C.1. Sustainability vs income inequality

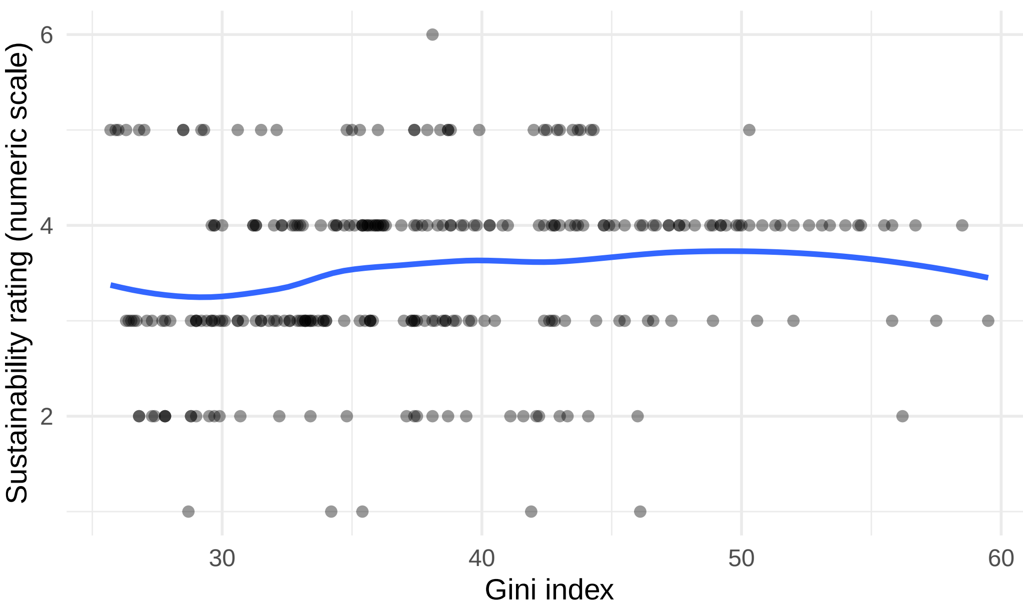
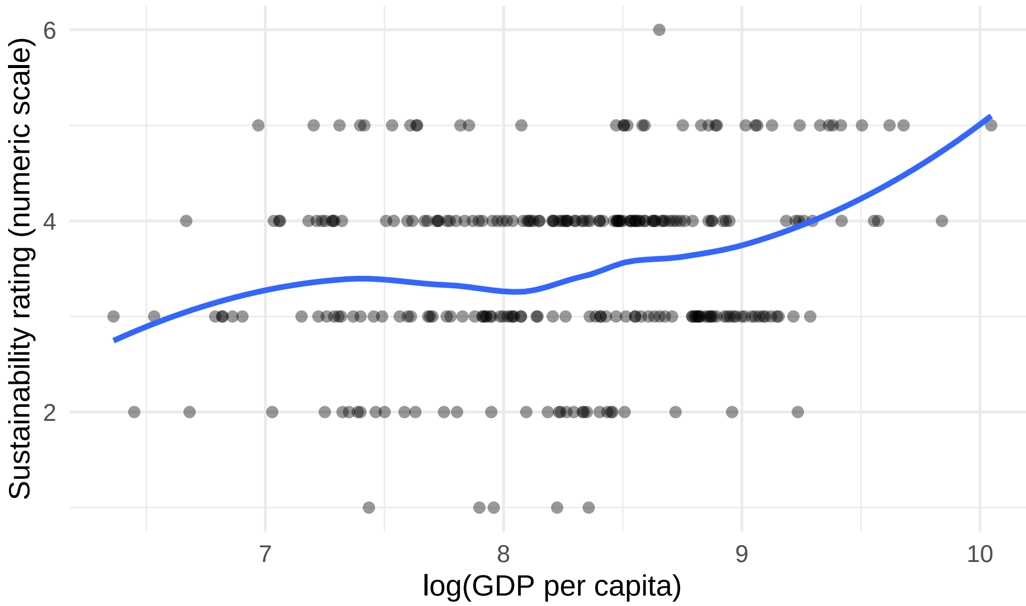


Figure C.2. Sustainability vs log(GDP) per capita




```
Testing proportional odds assumption by fitting separate models at each threshold:
```

Table 14: Coefficients at Each Threshold (Proportional Odds Check)

cutpoint	Gini	log_GDP
2	0.032	0.733
2.5	0.102	1.349
3	0.121	0.926
3.5	-0.011	0.631
4	0.012	2.355

```
If proportional odds assumption holds, coefficients should be similar across thresholds.
```

Brief Interpretation:

These plots support the idea that:

- Sustainability ratings increase with log(GDP) in a roughly monotonic way.
- The relationship with Gini index is weaker but still positive.

We rely on the standard proportional odds assumption for interpretation, and there is no obvious violation in the ordered pattern of predicted probabilities across categories. Both inequality and GDP maintain positive associations across all category thresholds, with coefficient variation that is moderate and acceptable for this sample size.

Appendix D. Technical Next Steps

This project is well-structured with relatively sufficient data, but is still mainly constrained by data availability, measurement choices, and the scope of methods. Below we summarize key limitations and potential next steps for improving the analysis. These steps would refine, rather than overturn, the main conclusions.

D.1 Data refinement

1. **Filling gaps in inequality data (Gini)**

   - The Gini index series is reasonably broad, but there are still missing country–years, especially for low-income countries and earlier periods. This leads to dropping some observations when constructing the panels.
   - **Next step:** supplement the current Gini index series with other harmonized sources (for example SWIID or LIS) where available and test whether the main coefficients change when the panel becomes denser. This would mainly improve precision and robustness rather than fundamentally changing the modelling strategy.

2. **Additional covariates as sensitivity checks**

   - The current models focus on the main story (inequality, economic productivity, and outcomes) plus time and region controls. Other plausible confounders (education, health spending, governance) are not included, mainly to keep the models interpretable.
   - **Next step:** run supplementary models that add a small set of extra controls where data are available. If results remain similar, this would strengthen confidence that the main findings are not driven by omitted-variable bias.

D.2 Modelling refinements

1. **Trying alternative panel structures for life expectancy**

   - The current linear models with year fixed effects already capture much of the variation in life expectancy. However, they treat country differences largely through region and not through explicit country effects.

- **Next step:** estimate a country fixed-effects or random-effects version of the life expectancy model as a robustness check. Comparing these estimates with the existing model would show whether within-country changes over time tell a similar story about inequality, income, and life expectancy, without replacing the current specification as the main result.

2. Exploring mild non-linearity in Gini index and log(GDP)

- Using log(GDP) already improves linearity and residual patterns. There may still be gentle curvature, especially at very high or low levels of income or inequality.
- **Next step:** add simple spline terms or quadratic terms for log(GDP) and/or Gini index in secondary models. If these terms are small, we can report that the linear approximation is adequate; if they matter, we can describe the relationship as slightly non-linear while still keeping the interpretation straightforward.

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